Emotion Detection Mastery: EEG Signal Analysis with Balanced vs. Unbalanced Data

Jadov Menaka
School of Data Science
Indian Institute of Science Education and Research
Thiruvananthapuram - 695551, Kerala, India
jadovmenaka20@iisertvm.ac.in

Sreelekshmi I G
Department of Computational Biology and Bioinformatics
University of Kerala
Karyavattom, Thiruvananthapuram - 695581, Kerala, India
sreelekshmimeenu1999@gmail.com

Alwin Poulose*
School of Data Science
Indian Institute of Science Education and Research
Thiruvananthapuram - 695551, Kerala, India
alwinpoulosepalatty@iisertvm.ac.in

2024 International Conference on Brain Computer Interface & Healthcare Technologies (iCon-BCIHT) | 979-8-3315-4006-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICON-BCIHT63907.2024.10882443

***Abstract*—Emotion is crucial in daily life, influencing stress levels, decision-making, relationships, and overall mental health. Recognizing emotions accurately is valuable across sectors like healthcare, education, and beyond. This paper presents a comparative study on emotion recognition, applying various deep learning models with the DEAP dataset, which includes EEG recordings from 32 participants who watched 40 short music video clips. We extracted time-domain, frequency-domain, and wavelet features from the preprocessed EEG data to capture nuanced patterns within the signals. The models assessed in this study include convolutional neural networks (CNN), long short-term memory networks (LSTM), gated recurrent units (GRU), and hybrid architectures, each trained on both balanced and unbalanced datasets. To address class imbalance, data was augmented using the Synthetic Minority Oversampling Technique (SMOTE). Results indicate that the Attention GRU model achieved slightly higher performance by utilizing its attention mechanism to capture extended dependencies within the data. Additionally, model accuracy improved by 5-9% when trained on balanced datasets, underscoring the importance of data balancing in enhancing classification outcomes.**

***Keywords*—Emotion Recognition; Electroencephalogram (EEG); DEAP Dataset; Synthetic Minority Oversampling Technique (SMOTE); Deep Learning Models.**

I. Introduction

Emotion is a physiological response to external stimuli and internal cognitive processes involving a series of physiological activities in the human body. Accurate understanding and recognition of emotions are crucial for various applications, including learning systems, mental health assessment, and human-computer interaction (HCI). Emotion recognition has emerged as a significant research area due to its numerous applications, such as monitoring driver emotions, enhancing healthcare, and improving entertainment experiences. Different data modalities are used for emotion recognition, including behavioural and physiological signals. Physiological signals, such as Human Rate Variability (HRV), Electroencephalography (EEG), Electrocardiogram (ECG), Electromyography (EMG), Respiration Rate (RR), and ElectroDermal Activity (EDA), represent internal responses to emotional stimuli, providing insights into a person's physiological state. Among these methods, EEG stands out due to its ability to directly measure brain activity, offering a more objective understanding of emotional states. Emotion recognition using EEG signals combines neuroscience and artificial intelligence, opening new frontiers in affective computing.

This study focuses on emotion recognition using the DEAP dataset, which includes EEG and other physiological signals recorded from 32 participants after watching 40 video clips. The primary aim is to identify the best model and conduct a comparative study of proposed models on both balanced and unbalanced data. The study involves preprocessing and extracting time-domain features (e.g., Hjorth parameters, zero crossing rate, waveform length), frequency-domain features (e.g., power spectral density, spectral entropy, spectral centroid, spectral bandwidth, band power, relative power), and wavelet features (e.g., wavelet entropy) from EEG signals. Various deep learning models were implemented, including convolutional neural networks (CNN), gated recurrent units (GRU), long short-term memory (LSTM), BiLSTM, BiGRU, Attention LSTM, and Attention GRU. These models were applied to balanced and unbalanced data, and their performance was compared. The comparisons suggest an increase in model accuracy when transitioning from unbalanced to balanced data using the Synthetic Minority Over-Sampling Technique (SMOTE) algorithm. This research aims to provide a comprehensive comparative analysis of deep learning models on the DEAP dataset, facilitating future studies to achieve prompt and precise analyses.

II. Related Work

Research on emotion recognition has significantly grown, with applications ranging from improving HCI to advancing healthcare. Recognizing and understanding emotions is increasingly vital in education, entertainment, and mental health.

979-8-3503-8168-9/24/$31.00 ©2024 IEEE

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 19,2026 at 09:46:53 UTC from IEEE Xplore. Restrictions apply.

Recently, using EEG signals to study intricate patterns has revolutionized emotion recognition. Researchers have focused on identifying EEG patterns corresponding to various emotional states using advanced signal processing techniques, machine learning, and deep learning algorithms to enhance performance. This literature review presents an overview of state-of-the-art emotion recognition utilizing EEG signals.

The DEAP dataset, introduced by the authors in [1], is a benchmark for emotion recognition research. Using a valence-arousal scale, they proposed a music video recommendation system that translates users' physiological responses into emotions. The dataset includes EEG and peripheral physiological signals recorded from participants exposed to emotional stimuli, providing a comprehensive statistical analysis and preliminary classification results. The authors in [2] survey critical aspects of emotion recognition, including inter-subject data variance, data annotation techniques, preprocessing methods, and multimodal fusion approaches. Their work provides a comprehensive comparison of different emotion recognition models and strategies. An in-depth analysis using DEAP and SEED datasets is presented in [3], focusing on data collection, preprocessing, and interpretation of EEG signals. This study employs machine learning and signal processing techniques to explore EEG signals for emotion recognition, achieving an arousal accuracy of 70.88% and an overall accuracy of 76.00% using an SVM classifier. In [4], the authors propose hybrid models combining CNN-1D and RNN architectures, specifically GRU and LSTM, to classify emotional states on the DEAP dataset. Their models achieve high training accuracy, demonstrating the effectiveness of hybrid approaches in emotion classification tasks. The authors in [5] address challenges in emotion recognition, such as capturing correlations between physiological signals over time and minimizing the effects of Temporal Covariate Shift (TCS). They propose using residual networks and domain generalization techniques, validated on the WESAD and DEAP datasets, to enhance emotion recognition from physiological data. The work in [6] proposes three solutions for emotion recognition: a hybrid CNN-RNN model, a contrastive GAN-based Graph Neural Network, and a transformer-based bimodal model using EEG and facial expressions. These approaches, applied to the DEAP and MAHNOB datasets, highlight the potential of advanced neural network architectures in emotion recognition. The research in [7] presents an intelligent emotion detection system utilizing a hybrid stacked GRU and RNN architecture. They employ the SMOTE-Tomek method for class balancing and PCA for feature optimization, achieving superior performance on the AMIGOS and DEAP datasets.

This literature review underscores recent advancements in emotion recognition using EEG signals, showcasing various applications and methodologies. Researchers have proposed numerous signal processing, machine learning, and deep learning methods to enhance emotion recognition accuracy. Notable contributions include addressing challenges like inter-subject data variance and artifact removal. Future research should focus on overcoming these limitations and exploring multimodal approaches to enhance emotion recognition robustness across various domains.

III. Methodology

The methodology used in our study is illustrated in Fig. 1, which includes the DEAP dataset, data preprocessing, feature extraction, labelling, deep learning (DL) models and model evaluation.

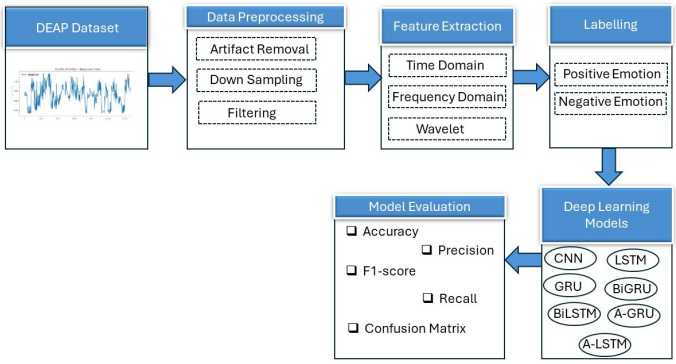


Fig. 1: System Overview.

A. DEAP Dataset

This study employed the multimodal DEAP dataset to examine different emotional states. Recordings were made using the International 10-20 system, a widely accepted approach in EEG research, to ensure standardized and repeatable electrode placements. The dataset includes EEG and peripheral signals from 32 participants, each exposed to 40 one-minute music video clips. Participants rated each clip on arousal, valence, like/dislike, dominance, and familiarity. Both raw and preprocessed data are available in the dataset; however, this study utilized the preprocessed version, which includes data that has been downsampled to 128 Hz, preprocessed, and segmented, available in pickled/numpy formats. Each participant's file contains two arrays: a data array and a label array. The data array has 40 channels, with the first 32 dedicated to EEG data and the remaining channels representing additional peripheral signals. The label array provides four emotion-related labels: valence, arousal, dominance, and liking.

B. Data Preprocessing

The EEG data collected from the DEAP dataset were first preprocessed using artifact removal, downsampling, and band-pass filtering methods.

1) Artifact Removal: The significance of signal artifacts increases when gathering EEG data, potentially compromising data quality. Artifacts, or unwanted signals, originate from experimental error, environmental noise, and physiological sources such as EOG, ECG, and EMG. Extrinsic artifacts stem from external sources like experimental errors and environmental noise, while intrinsic artifacts come from the body itself, including heart rate, muscle contraction, and eye blink.

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 19,2026 at 09:46:53 UTC from IEEE Xplore. Restrictions apply.

2) Downsampling: Downsampling is employed in preprocessing to reduce the sampling rate of EEG signals, thereby decreasing computational power requirements while preserving essential information. As per the Nyquist theorem, EEG signals are typically recorded at high sampling rates to capture all necessary frequency components depicting brain activity. However, these high sampling rates generate substantial data, complicating computer analysis and processing. Downsampling mitigates this issue by lowering the sampling rate, thus making the data more manageable by reducing its size.

3) Filtering: Filtering involves removing specific frequency components from recorded EEG signals to improve signal quality and facilitate better analysis. Bandpass filtering, a commonly used method, permits only signals within a specified frequency range while attenuating those outside it, thereby preserving the relevant frequency components of EEG signals. In this study, a bandpass filter ranging from 4.0 to 45.0 Hz was applied, covering all pertinent frequency bands.

C. Feature Extraction

Feature extraction is a critical step in preprocessing, as it isolates and removes meaningful information from raw data for effective analysis. For accurate emotion recognition using EEG signals, features can be extracted from various domains, including the time, frequency, and wavelet domains, each capturing distinct patterns in the EEG signals.

In the time domain, features are derived from EEG signals over time, revealing how the signals change. Notable time-domain features include Hjorth parameters, zero crossing rate, skewness, kurtosis, and waveform length. These features capture essential characteristics of EEG data, aiding in distinguishing various emotional states.

Frequency domain features are obtained by transforming the EEG signal into the frequency domain, typically using the Fast Fourier Transform (FFT). FFT decomposes the EEG data into individual frequencies, indicating the signal's frequency content. This transformation facilitates the analysis and extraction of the signal's spectral content. Key frequency domain features include Power Spectral Density, Spectral Centroid, Spectral Entropy, Spectral Bandwidth, Band Power, Relative Power, Peak Frequency, and Harmonic Ratio.

Wavelet features are derived using wavelet transform, a robust tool representing an EEG signal in both time and frequency domains. The wavelet transform decomposes the signal into multiple resolution levels corresponding to different frequency bands, allowing for the examination of the signal at various resolutions and capturing its high- and low-frequency components. Notable wavelet features include wavelet entropy and wavelet energy.

D. Labelling

This study's labelling is based on arousal values with a threshold of 5. If the arousal value exceeds 5, the label is 'Positive'; otherwise, it is 'Negative'. Arousal values were explicitly chosen for labelling due to their critical role in differentiating levels of emotional intensity. Arousal measures indicate psychological activity or alertness and directly indicate how strongly an individual will react to a given stimulus. Unlike other emotional dimensions, such as valence (an emotion's positive or negative quality), arousal is more readily indexed by EEG measures and frequently exhibits clear patterns in brain activity. Focusing on arousal provides more precise and interpretable EEG patterns, enhancing the accuracy and reliability of emotion recognition.

E. Deep Learning Models

1) CNN: A CNN is an advanced form of an Artificial Neural Network (ANN) specifically designed to extract features from grid-like matrix datasets, such as images and videos . It comprises several layers: input, convolutional, pooling, fully connected, and output. The convolutional layers detect features in the input, such as edges, textures, and patterns, by applying filters. Each filter slides over the input, performing element-wise multiplication and summation to produce a feature map. Pooling layers then reduce the dimensionality of the feature maps, mitigating the risk of overfitting. Finally, the fully connected layers integrate all the features detected by the previous layers to make a final decision.

2) LSTM: An LSTM network is a type of Recurrent Neural Network (RNN) that effectively processes sequential data, such as text, audio, or time series. LSTM networks can retain information for extended periods, which is crucial for understanding context in sequences. This capability is achieved through a memory cell mechanism. The LSTM cell operates through three key regions, known as gates, during each timestep: the forget gate, the input gate, and the output gate. The forget gate decides whether the information from the previous timestep is retained or discarded based on its relevance. The input gate determines the new information to be learned from the current input. The output gate updates and passes the information to the next time step. Additionally, the LSTM network maintains a hidden state, denoted as H_{t-1} for the previous timestep and H_t for the current timestep, as well as memory cells C_{t-1} and C_t for the previous and current timesteps, respectively.

3) GRU: A GRU is a RNN type designed to address the vanishing gradient problem and effectively handle long-term dependencies. This makes GRUs particularly useful for tasks such as predicting the next word in a sentence or the subsequent value in a time series. GRUs consist of two primary gates: an update gate and a reset gate. The update gate determines the extent to which past memory should be retained. The Reset gate controls how much of the past memory should be forgotten when forming a new candidate memory. The candidate memory is a draft of the new memory, created based on new information and modified past memories. Unlike LSTM networks, GRUs maintain short- and long-term dependencies within a single hidden state. The GRU's gating mechanism allows the hidden state to simultaneously manage long-term and short-term dependencies, streamlining

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 19,2026 at 09:46:53 UTC from IEEE Xplore. Restrictions apply.

the computations and the overall architecture compared to LSTMs.

4) Bidirectional Models: A bidirectional model is a neural network architecture that processes input data in both forward and backward directions, enabling it to capture contextual information from both past and future states. This capability is particularly advantageous when handling sequential data, as the bidirectional approach enriches the understanding of the data by incorporating information from both directions. This mechanism is highly beneficial in natural language processing (NLP) and time series analysis. A bidirectional mechanism can be applied to LSTM networks and GRUs. In a bidirectional setup, the input sequences are processed by two RNNs: the forward RNN, which processes the data from the beginning to the end, and the backward RNN, which processes the data from the end to the beginning. The outputs of these two RNNs are then combined to generate the final output, thereby enhancing the model's performance by leveraging the comprehensive contextual information from both directions.

5) Attention Mechanism: The attention mechanism is a crucial technique in neural networks, especially for sequence modeling tasks such as machine translation and text summarization. By assigning importance to specific parts of the input sequence, it enhances the model's performance. The attention mechanism operates within an encoder-decoder architecture, allowing the model to focus on relevant segments of the input during task execution. This is achieved by dynamically allocating weights to different parts of the input, indicating their relative importance or relevance. By processing and selectively attending to the most pertinent information, the model can identify dependencies and relationships within the data, leading to improved performance and accuracy.

F. Model Evaluation

To comprehensively assess the effectiveness of the deep learning models, we evaluated their performance using several key metrics: accuracy, precision, recall, F1 score, and the confusion matrix. Together, these metrics provide a thorough framework to gauge the models' accuracy and reliability in recognizing emotions based on EEG signals.

IV. Results and Discussion

This study applied several deep learning architectures to the dataset, including CNN, LSTM, GRU, BiLSTM, BiGRU, and attention-based LSTM and GRU models (A-LSTM and A-GRU). The CNN model architecture consisted of two Conv1D layers, each with 32 filters, a kernel size of 3, and the same padding, followed by a max-pooling layer with a pool size of 2. A dropout layer with a rate of 0.5 was placed after the max-pooling layer. Two dense layers followed, with the first dense layer containing 64 units. The Rectified Linear Unit (ReLU) activation function was used in the convolution and pooling layers. In contrast, a softmax function was applied in the output layer for classification, expecting the target labels to be one-hot encoded. The LSTM model included one LSTM layer with 256 units, followed by two dropout layers and two dense layers, the first with 128 units. ReLU was used in the LSTM layers. The GRU model architecture consisted of two GRU layers, each with 256 units, followed by a dropout layer and a dense layer. The BiLSTM model had two LSTM layers, with 256 units in the first and 128 units in the second, followed by two dropout layers (rate of 0.5) and two dense layers. Similarly, the BiGRU model included two GRU layers of 256 units each, along with two dropout layers (rate of 0.5) and two dense layers. The LSTM with an attention mechanism contained two LSTM layers, with 256 units in the first layer and 128 units in the second, followed by two dropout layers and two dense layers. The attention-based GRU model (A-GRU) had two GRU layers, with the first layer containing 256 units, followed by two dropout layers and two dense layers. Each model was implemented and trained using K-Fold Cross Validation to evaluate their performance on the emotion recognition task using the DEAP dataset.

For all models, the ReLU activation function was applied throughout the layers. In contrast, the output dense layer employed a softmax activation function to predict class probabilities, with target labels prepared in a one-hot encoded format. This setup allows the models to assign probabilities to each class effectively. Additionally, a dropout layer with a rate of 0.5 followed the dense layers to address potential overfitting and improve generalization. The Adam optimizer was chosen for model optimization due to its efficiency and adaptability. Categorical cross-entropy served as the loss function was suitable for multi-class classification tasks and aligned well with the objectives of this study. Each model was trained over 50 epochs with a batch size 32 to ensure a comprehensive learning process. The model training was conducted on balanced and unbalanced datasets, summarized in TABLES I and II.

TABLE I: Deep learning models classification performance on unbalanced data.

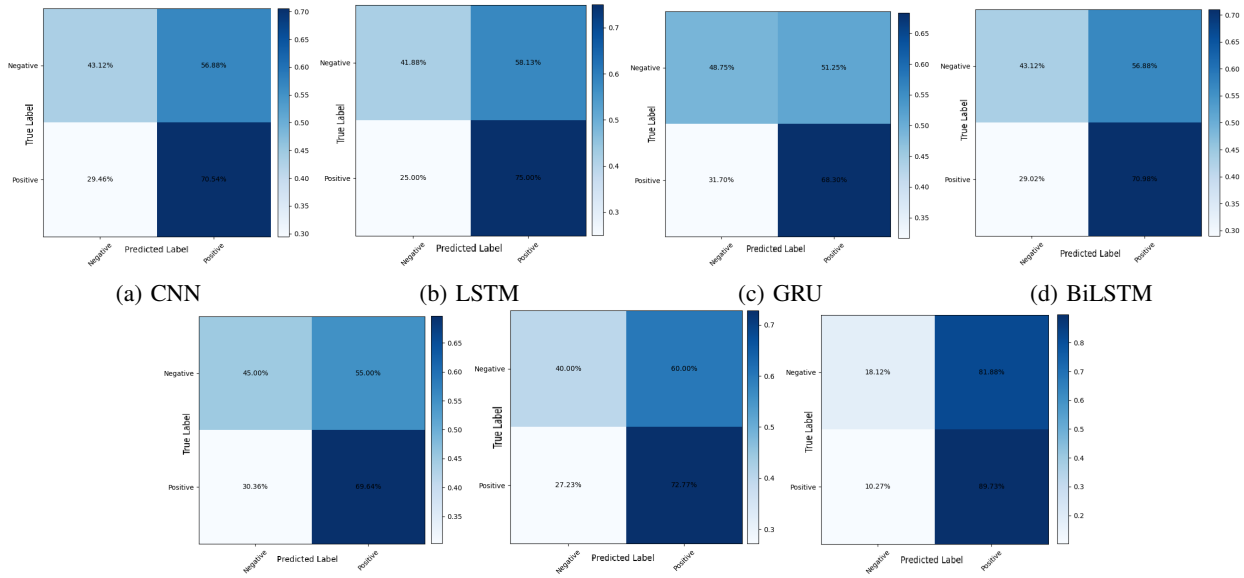
Model	Cross Validation Accuracy	Precision	Recall	F1 score
CNN	58%	57%	57%	57%
LSTM	60%	59%	580%	58%
GRU	59%	59%	59%	59%
BiLSTM	59%	58%	57%	57%
BiGRU	58%	58%	57%	57%
A-LSTM	58%	57%	56%	56%
A-GRU	62%	58%	54%	50%

TABLE II: Deep learning models classification performance on balanced data.

Model	Cross Validation Accuracy	F1 score	Precision	Recall
CNN	66%	66%	66%	66%
LSTM	67%	67%	67%	67%
GRU	66%	66%	66%	66%
BiLSTM	67%	67%	67%	66%
BiGRU	67%	67%	67%	67%
A-LSTM	66%	66%	66%	66%
A-GRU	66%	66%	66%	66%

The results for the unbalanced data, as shown in TABLE I, indicate that the CNN model achieved an accuracy of 58%,

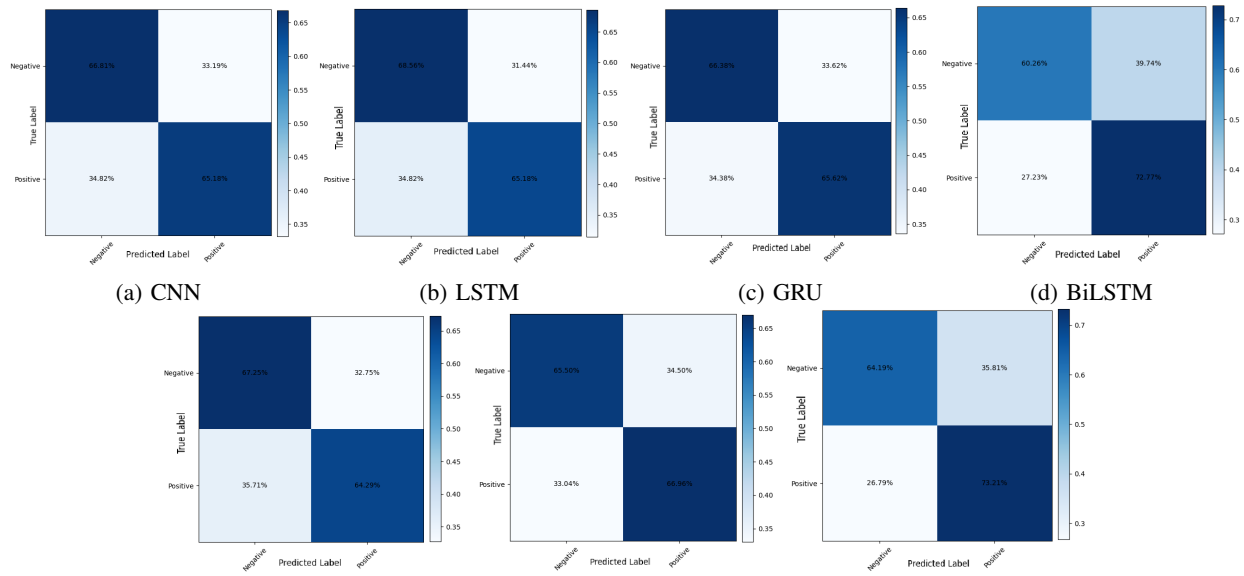
Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 19,2026 at 09:46:53 UTC from IEEE Xplore. Restrictions apply.



(a) CNN (b) LSTM (c) GRU (d) BiLSTM

(e) BiGRU (f) A-LSTM (g) A-GRU

Fig. 2: Confusion matrix results of deep learning models on unbalanced data.



(a) CNN (b) LSTM (c) GRU (d) BiLSTM

(e) BiGRU (f) A-LSTM (g) A-GRU

Fig. 3: Confusion matrix results of deep learning models on balanced data.

LSTM 60%, GRU 59%, BiLSTM 59%, BiGRU 58%, Attention LSTM 58%, and Attention GRU 62%. Among these, the Attention GRU model exhibited slightly superior performance compared to the other models. This enhanced performance can be attributed to the attention mechanism, which improves the model's ability to focus on crucial parts of the input sequence, thereby capturing long-range dependencies. For the balanced data, summarized in TABLE II, the accuracy results were as follows: CNN 66%, LSTM 67%, GRU 66%, BiLSTM 67%, BiGRU 67%, Attention LSTM 66%, and Attention GRU 66%. The LSTM, BiLSTM, and BiGRU models performed marginally better than the others in this comparison. When comparing the performance of balanced and unbalanced data, we observed an increase in accuracy and performance of 4% to 9% with the balanced data.

Furthermore, the performance of the models was analyzed using confusion matrix results, as illustrated in Figs. 2 and 3, which present the outcomes for both balanced and unbalanced data. The CNN model demonstrated notable accuracy in predicting various emotions, with 43% accuracy for negative emotions, 71% for positive emotions on unbalanced data, 67% for negative emotions and 65% for positive emotions on balanced

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 19,2026 at 09:46:53 UTC from IEEE Xplore. Restrictions apply.

data. The LSTM model showed 42% accuracy for negative emotions, 75% for positive emotions on unbalanced data, 69% for negative emotions and 65% for positive emotions on balanced data. The GRU model achieved 49% accuracy for negative emotions, 68% for positive emotions on unbalanced data, and 66% for negative and positive emotions on balanced data. The BiLSTM model predicted the negative emotions with 43% accuracy and positive emotions with 71% accuracy on unbalanced data, 60% for negative emotions and 73% for positive emotions on balanced data. The BiGRU model showed 45% accuracy for negative emotions, 70% for positive emotions on unbalanced data, 73% for negative emotions and 63% for positive emotions on balanced data. The Attention LSTM model recorded negative emotions with 40% accuracy and positive emotions with 73% accuracy on unbalanced data, 66% for negative emotions and 67% for positive emotions on balanced data. The Attention GRU model scored 18% accuracy for negative emotions, 90% for positive emotions on unbalanced data, 64% for negative emotions and 73% for positive emotions on balanced data.

This comparative result analysis shows that most negative classes were mispredicted on the unbalanced data but were correctly predicted on the balanced data. Additionally, the accuracy of the models improved when trained on balanced data compared to unbalanced data. This indicates that balancing the data improves the models' performance, making them more effective and reliable in capturing essential features from EEG signals for emotion recognition.

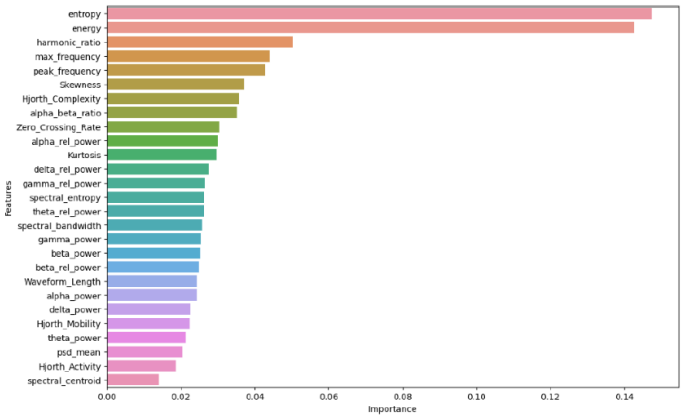


Fig. 4: Feature importance graph.

The importance of features within the DEAP dataset was further examined to validate their impact on the performance of deep learning models. Fig. 4 illustrates the feature importance graph, which provides insights into the contribution of each feature to the classification task. The graph highlights that wavelet-based features, such as wavelet entropy and wavelet energy, play a critical role in feature extraction. This is largely due to their capacity to capture information across both the time and frequency domains, which is essential for identifying nuanced patterns within EEG signals. Conversely, the spectral centroid emerged as one of the least influential features. While it offers insights into the frequency domain, its contributions are somewhat redundant, as it largely overlaps with information provided by other frequency-related features. The spectral centroid is also limited in capturing the more complex and intricate signal patterns essential for accurate emotion recognition. This suggests that while frequency domain features contribute to model accuracy, certain features like wavelet energy and entropy are more efficient in providing the comprehensive representation needed for robust classification performance.

V. Conclusion

In this paper, we utilized the DEAP dataset for emotion recognition, employing various deep-learning models and conducting a comparative analysis of balanced and unbalanced data. We extracted and combined multiple features from EEG signals. Using K-Fold cross-validation to assess performance, we trained the dataset on CNN, LSTM, GRU, BiLSTM, BiGRU, Attention LSTM, and Attention GRU models. The unbalanced data was resampled and converted to balanced data using the SMOTE algorithm to oversample the minority class and improve model performance. Our experiments demonstrated that the Attention GRU model slightly outperformed others in the unbalanced class due to its attention mechanism enhancing its focus on essential parts of the input sequence and capturing long-range dependencies. Despite these promising results, our study has several limitations, including using only a single dataset, which may restrict the generalization of our findings. Future research should aim to validate these models on more extensive and diverse EEG datasets to enhance generalizability and explore advanced deep learning models and transformer models to increase the robustness of emotion recognition.

Acknowledgment

We thank Indian Institute of Science Education and Research Thiruvananthapuram (IISER TVM) for providing the facilities needed for this paper's successful completion.

References

[1] S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, and I. Patras, "Deap: A database for emotion analysis; using physiological signals," *IEEE transactions on affective computing*, vol. 3, no. 1, pp. 18–31, 2011.

[2] Z. Ahmad and N. Khan, "A survey on physiological signal-based emotion recognition," *Bioengineering*, vol. 9, no. 11, p. 688, 2022.

[3] S. K. Jha, S. Suvvari, and M. Kumar, "Eeg-based emotion recognition: An in-depth analysis using deap and seed datasets," in *2024 11th International Conference on Computing for Sustainable Global Development (INDIACom)*. IEEE, 2024, pp. 1816–1821.

[4] F. Zamani, R. Wulansari *et al.*, "Emotion classification using 1d-cnn and rnn based on deap dataset," *Nat. Lang. Process*, pp. 363–378, 2021.

[5] J. Li, J. Li, X. Wang, X. Zhan, and Z. Zeng, "A domain generalization and residual network-based emotion recognition from physiological signals," *Cyborg and Bionic Systems*, vol. 5, p. 0074, 2024.

[6] S. Soleimani, "Deep learning architectures for enhanced emotion recognition from eeg and facial expressions," Ph.D. dissertation, Université d'Ottawa/University of Ottawa, 2024.

[7] D. Nandini, J. Yadav, A. Rani, and V. Singh, "Enhancing emotion detection with non-invasive multi-channel eeg and hybrid deep learning architecture," *Iranian Journal of Science and Technology, Transactions of Electrical Engineering*, pp. 1–20, 2024.

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 19,2026 at 09:46:53 UTC from IEEE Xplore. Restrictions apply.